

Name:

SOLNS

Date: _____



Maths Year 9

Investigation 3, 2011

Topic - Binary Numbers

Important Information:

Although the take-home component is not worth any marks, it is essential in preparation for the in-class component. Knowledge and skills gained will be extended in the in-class validation component. This in-class validation will be completed under test conditions on the day in which this take-home component is due. The take-home component may be used when completing the in-class component. Contact will be made to parent(s) if the take-home component is not available for submission (at the start of the lesson).

Date out: Week 5 Date _____ Date Due: Week 6 Date _____

Take home component weighting: 0% of the year In-class component weighting: _____ % of the year.

AIM:

In this assessment, you will be investigating Binary numbers.

There are only 10 types
of people in the world:
Those who understand binary
and those who don't.

Does the above statement make sense?

The binary number system or base-2 number system is a way of representing numbers using two symbols 0 and 1. As this can be directly be used to represent electrical switches (which are either on or off) the binary system is used internally in most computers.

Our number system is base-10, it uses ten digits to write numbers and each place value increases by a factor of 10.

Consider the number 35792

Digit	3	5	7	9	2
Place value	10,000 (ten thousands)	1000 (thousands)	100 (hundreds)	10 (tens)	1
Place value as an indice	10^4	10^3	10^2	10^1	10^0

We can re-write 35792 in expanded notation as $3 \times 10000 + 5 \times 1000 + 7 \times 100 + 9 \times 10 + 2 \times 1$

or in expanded notation with the place values in index form

$$3 \times 10^4 + 5 \times 10^3 + 7 \times 10^2 + 9 \times 10^1 + 2 \times 10^0$$

1. Write the following numbers in expanded notation using index form

a) 342

$$3 \times 10^2 + 4 \times 10^1 + 2 \times 10^0$$

b) 6592

$$6 \times 10^3 + 5 \times 10^2 + 9 \times 10^1 + 2 \times 10^0$$

c) 95276

$$9 \times 10^4 + 5 \times 10^3 + 2 \times 10^2 + 7 \times 10^1 + 6 \times 10^0$$

3 Complete the table for the following binary numbers

a) 11101_2

Digit	1	1	1	0	1
Place value	16	8	4	2	1
Place value as an indice	2^4	2^3	2^2	2^1	2^0

$$\begin{aligned}
 11101_2 &= 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \quad (\text{expanded notation index form}) \\
 &= 1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 \quad (\text{expanded notation}) \\
 &= 31
 \end{aligned}$$

So 11101_2 is the number 31

b) 110010_2

Digit	1	1	0	0	1	0
Place value	32	16	8	4	2	1
Place value as an indice	2^5	2^4	2^3	2^2	2^1	2^0

$$\begin{aligned}
 110010_2 &= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \quad (\text{expanded notation index form}) \\
 &= 1 \times 32 + 1 \times 16 + 0 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 \quad (\text{expanded notation}) \\
 &= 50
 \end{aligned}$$

So 11010_2 is the number 50

c) 1010110_2

Digit	1	0	1	0	1	1	0
Place value	64	32	16	8	4	2	1
Place value as an indice	2^6	2^5	2^4	2^3	2^2	2^1	2^0

$$\begin{aligned}
 1010110_2 &= 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \quad (\text{expanded notation index form}) \\
 &= 1 \times 64 + 0 \times 32 + 1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 \quad (\text{expanded notation}) \\
 &= 86
 \end{aligned}$$

So 111012 is the number 86

The binary system is base-2, it uses two digits 0 and 1 to write numbers and each place value increases by a factor of 2. To distinguish a binary number from a normal number (base ten) we write a small 2 after it.

Eg

101 is one hundred and one

10101₂ is a binary number

Binary number 11010₂

Digit	1	1	0	1	0
Place value	16	8	4	2	1
Place value as an indice	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

$$\begin{aligned}
 11010_2 &= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 && \text{(expanded notation index form)} \\
 &= 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 && \text{(expanded notation)} \\
 &= 26
 \end{aligned}$$

So 11010₂ in binary is actually the number 26

2 Complete the following table of the first ten binary numbers

Binary number	Expanded notation index form	Expanded notation	Base-10 number
1 ₂	1 x 2 ⁰	1 x 1	1
10 ₂	1 x 2 ¹ + 0 x 2 ⁰	1 x 2 + 0 x 1	2
11 ₂	1 x 2 ¹ + 1 x 2 ⁰	1 x 2 + 1 x 1	3
100 ₂	1 x 2 ² + 0 x 2 ¹ + 0 x 2 ⁰	1 x 4 + 0 x 2 + 0 x 1	4
101 ₂	1 x 2 ² + 0 x 2 ¹ + 1 x 2 ⁰	1 x 4 + 0 x 2 + 1 x 1	5
110 ₂	1 x 2 ² + 1 x 2 ¹ + 0 x 2 ⁰	1 x 4 + 1 x 2 + 0 x 1	6
111 ₂	1 x 2 ² + 1 x 2 ¹ + 1 x 2 ⁰	1 x 4 + 1 x 2 + 1 x 1	7
1000 ₂	1 x 2 ³ + 0 x 2 ² + 0 x 2 ¹ + 0 x 2 ⁰	1 x 8 + 0 x 4 + 0 x 2 + 0 x 1	8
1001 ₂	1 x 2 ³ + 0 x 2 ² + 0 x 2 ¹ + 1 x 2 ⁰	1 x 8 + 0 x 4 + 0 x 2 + 1 x 1	9
1010 ₂	1 x 2 ³ + 0 x 2 ² + 1 x 2 ¹ + 0 x 2 ⁰	1 x 8 + 0 x 4 + 1 x 2 + 0 x 1	10

4 Complete the following table

Index	2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8
Value	1	2	4	8	16	32	64	128	256

To convert base ten numbers to binary we need to write the number as a sum of powers of two (the values from the bottom row in the table)

$$\begin{aligned}
 \text{Eg } 43 &= 32 + 8 + 2 + 1 \\
 &= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\
 &= 101011_2
 \end{aligned}$$

5 Complete the table for the numbers 11 to 20

Number	Number as a sum of powers of two	Sum in index form	Binary number
11	$8 + 2 + 1$	$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$	1011_2
12	$8 + 4$	$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$	1100_2
13	$8 + 4 + 1$	$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$	1101_2
14	$8 + 4 + 2$	$1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$	1110_2
15	$8 + 4 + 2 + 1$	$1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$	1111_2
16	16	$1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$	10000_2
17	$16 + 1$	$1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$	10001_2
18	$16 + 2$	$1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$	10010_2
19	$16 + 2 + 1$	$1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$	10011_2
20	$16 + 4$	$1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$	10100_2

5 Using the table of the previous page convert the following numbers into binary

a) 70

$$70 = 64 + \underline{4} + 2$$

$$= 1 \times \underline{2^6} + 0 \times \underline{2^5} + 0 \times \underline{2^4} + 0 \times \underline{2^3} + 0 \times 2^2 + \underline{1} \times \underline{2^1} + 0 \times \underline{2^0}$$

$$= 1000010_2 \quad (\text{binary number})$$

70 written in binary is 1000010₂

b) 38

$$38 = 32 + 4 + 2$$

$$= 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 100110_2 \quad (\text{binary number})$$

38 written in binary is 100110₂

c) 83

$$83 = 64 + 16 + 2 + 1$$

$$= 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= 1010011_2 \quad (\text{binary number})$$

38 written in binary is 1010011₂

d) 165

$$165 = 128 + \underline{32} + 4 + 1$$

$$= 1 \times 2^7 + 0 \times 2^6 + \underline{1} \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 10100101_2 \quad (\text{binary number})$$

38 written in binary is 10100101₂